

Race the Replay

Abstract

Baseball is entertainment, yet many of the game's most compelling stories remain invisible. Modern tracking data captures every movement and decision, but those insights rarely make their way to the fans watching from the stands. *Race the Replay* is a jumbotron game that gives baseball fans a way to engage and relive the biggest plays of a game. Using proprietary player and ball-tracking data with precise x/y coordinates recorded at millisecond intervals, *Race the Replay* identifies the most exciting plays from earlier in the game and brings them back to life on the scoreboard. Fans watch an animated field replay and guess how long it took a key runner to reach home plate, third base, or second base. Their guess locks in before the replay begins and then the correct time is revealed alongside a stadium-wide leaderboard.

Building this game required overcoming the constraints of an anonymized dataset, including the absence of player and team identities, play-by-play labels, and stadium dimensions. As a result, every replay and gameplay element had to be reconstructed from raw positional and event data alone. *Race the Replay* demonstrates how player-tracking technology can create interactive, data-driven entertainment, highlighting new opportunities to enhance the overall baseball fan experience.

Introduction/Background

Minor League Baseball has long served as a proving ground for both on-field talent and fan experience innovation. In recent years, ballparks across the country have invested in scoreboard technology, introducing features such as real-time pitch velocity, Automated

Ball-Strike (ABS) challenges, and spray charts during games. Despite these advances, most in-stadium data remains one-directional. Fans receive passive information, but they have few opportunities to interact with it. The need for greater engagement is particularly evident in Minor League Baseball. Research cited by Minor League Baseball found that families with children under 12 spend just two and a half innings actually watching a game they attend, leaving over six innings to fill with experiences beyond the action on the field (Maun, MiLB.com, 2017). As a result, many casual fans simply tune out the advanced statistics and visualizations displayed throughout the game. At the same time, player-tracking technology has transformed how baseball is analyzed. Systems capable of recording player and ball positions dozens of times per second now operate throughout professional baseball, generating detailed datasets used for scouting, player development, and strategic decision-making. Yet these insights have rarely made their way into experiences that engage fans inside the ballpark, leaving one of baseball's most valuable data assets largely absent from the fan experience. *Race the Replay* is built on the premise that bridging the gap between advanced tracking data and the live fan experience represents an untapped opportunity.

Efforts to bring tracking data closer to fans do exist at the Major League level. MLB's Statcast system made metrics like exit velocity, launch angle, and many others publicly accessible through platforms like Baseball Savant (baseballsavant.mlb.com). The NFL's Next Gen Stats platform (nextgenstats.nfl.com) has gone a step further, showing analytics live through positioning data shown directly in game broadcasts. Despite targeting different audiences, these platforms succeed because they translate complex data into stories that casual fans can easily understand. However, these tools are mostly broadcast-facing and consumption-based. They tell

fans what happened after the fact, but they do not invite fans to engage with the data during a live game. Traditional jumbotron games, such as trivia contests and dance cams, remain the most used form of in-stadium fan interaction. However, they are independent of the game itself and do not deepen fans' engagement with the action on the field. No widely adopted tool currently uses live player tracking data to power an interactive fan experience during a game. *Race the Replay* serves to fill that gap. By transforming the same positional tracking data used by analysts into a real-time guessing game displayed on the scoreboard, it creates a bridge between the front office analysts and the fan in seat 11 of section 108.

Data

Race the Replay is built on five interconnected tables provided as part of the SMT Data Challenge dataset, spanning 247 games of Minor League Baseball across one anonymized season. The first table records the ball's x/y/z coordinates at every moment of every play. The second records every player's x/y coordinates at the same moments, providing a continuous map of where each of the players on the field were at any given moment. The tracking tables record positions at millisecond intervals, meaning a single 15-second play can contain thousands of individual observations. In this coordinate system, home plate sits at the origin point of the field, with coordinates increasing toward first base along one axis and toward second base along the other. This coordinate system made it possible to calculate the locations of each base geometrically. The third table has discrete events during each play, such as when a pitch was thrown, when the ball was hit into play, when it deflected off a wall, and when play ended. Each event is tagged with a timestamp and a numeric code identifying what occurred. A lineups table

identifies which player occupied each fielding and baserunning position for every play. The game information table provides context about each game, including the anonymized date it was played and three-letter codes representing each team. While this table was not central to the replay game itself, it provided context for understanding the scope of the dataset and verifying that plays were drawn from a range of games.

Most importantly, the dataset is fully anonymized. Player names, team names, and stadium identities are replaced with arbitrary codes, and the data has no traditional play-by-play outcomes such as hits, outs, or runs. This anonymization shaped every step of how *Race the Replay* was built. There is no column that says a runner scored, and no field measuring how fast a player was moving. Every piece of information required to power the replay and guessing game had to be derived from scratch. Compelling plays were identified from patterns in the event data, base touches were detected by measuring the geometric distance between a runner's coordinates and the known location of each base, and player speed was calculated from changes in position between consecutive timestamps. Working within these constraints meant reconstructing information that would usually be available immediately from a box score or play-by-play feed. In a non-anonymized dataset, identifying when a runner scored is a single lookup. Here, it required geometry, event filtering, and millisecond-level position tracking all working together. Rather than limiting the project, reconstructing this missing information became its defining technical contribution.

Analysis

With no play-by-play labels or outcome data available, identifying which plays were worthy of replaying required developing a set of selection criteria. Raw play duration was the initial filter, but quickly proved unreliable because some of the longest plays in the dataset were irregular outliers rather than being long from on-field action. Instead, plays were filtered using signals directly tied to baseball “excitement.” The number of throws recorded during a play, whether the ball made contact with an outfield wall, and whether baserunners were confirmed to be in motion. Multiple throws combined with active baserunners often indicate defensive relays, close plays, and extended sequences that are more exciting for spectators. Using these filters, three plays were selected from across the dataset to represent distinct replay scenarios: 1) an extra-base hit with multiple runners in motion, 2) a ball deflecting off the outfield wall triggering a defensive relay, and 3) a close play at the plate that turned into a pickle (see Appendix A).

Once plays were selected, the next challenge was determining when each runner reached each base which is the answer fans are guessing. Because no explicit base-touch events existed in the dataset, arrival times had to be inferred geometrically. The known coordinates of each base were calculated from the field's coordinate system, and the distance between each runner's recorded position and each base was computed at every timestamp throughout the play. The moment a runner came within a defined threshold of a base was recorded as their arrival time (see Appendix A), and the elapsed time since the ball was hit was calculated to produce the final answer used in the game. Player speed was derived similarly. Runner velocity was estimated by measuring how far each player traveled between consecutive position readings and then dividing by the time elapsed between those readings. This produced a speed value in feet per second at every moment of the play (see Appendix A), which was then normalized to a zero-to-one scale

relative to a typical sprint speed in professional baseball. This normalized speed value is how the velocity trail displayed behind each runner in the animation was computed. Shifting from blue at slower speeds through yellow at mid-range effort to red at full sprint gives fans a visual read on the intensity of each moment.

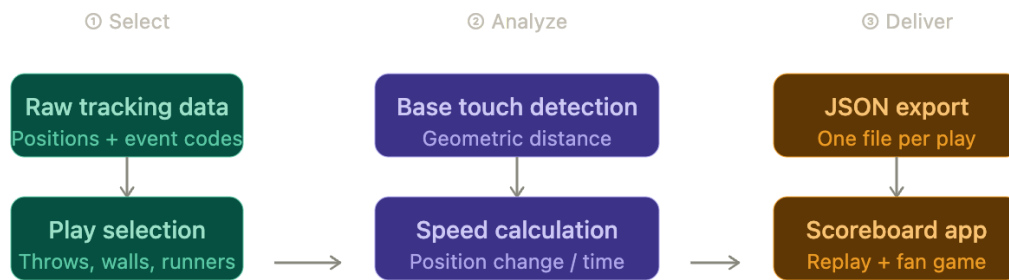


Figure 1. *Analytical pipeline from raw SMT tracking data to the Race the Replay scoreboard experience. Technical details for each step are provided in Appendix A.*

Full analysis code is submitted separately as part of the required submission package (see Appendix C).

Implementation

Race the Replay is designed to run on a jumbotron once or twice per game, triggered by a scoreboard operator in the later innings of a game to revisit the game's biggest plays from earlier.

The timing is deliberate so that fans have already experienced the moment live, making the replay and the guessing game feel like a natural recall. The experience unfolds in three “steps”.

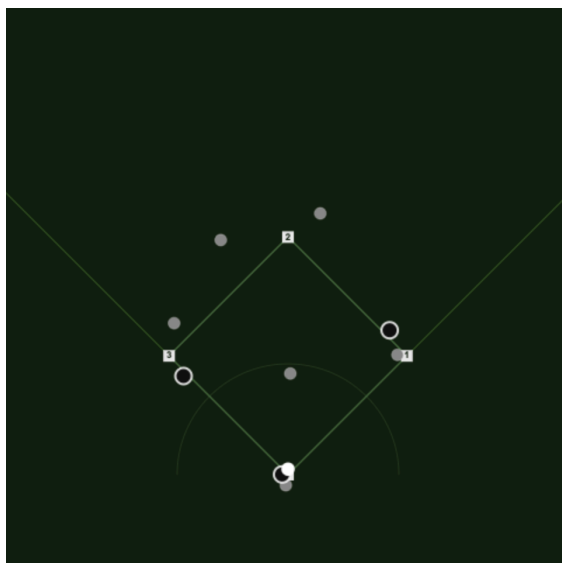


Figure 2. *Top-down field view at the start of a Race the Replay animation, showing offensive players (dark dots), defensive players (gray dots), and the ball (white dot) reconstructed from SMT player and ball tracking data.*

To start, an animated view of the baseball diamond appears on the scoreboard, made entirely from player and ball tracking data. Offensive players appear as dark dots with a velocity-colored trail that shifts from blue when decelerating to red at full sprint. This tool gives fans an immediate visual read on the hustle behind the play. Defensive players appear as smaller gray dots, providing field context without pulling too much attention away from the runners. The ball appears as a white dot that grows slightly when airborne, reflecting its height above the field.

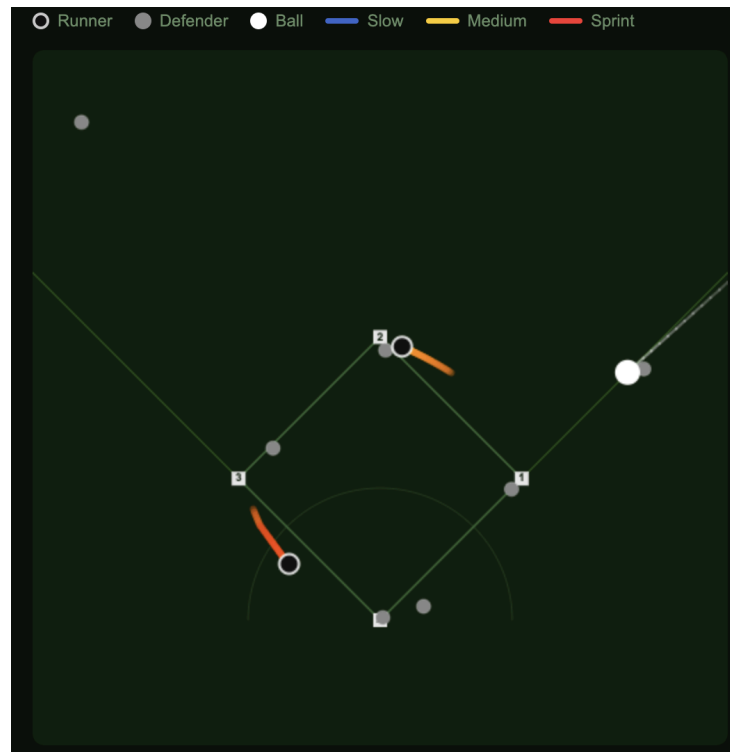


Figure 3. *Mid-play view with velocity-colored movement trails visible behind offensive players. Trail color shifts from blue (decelerating) through yellow (moderate effort) to red (full sprint), derived from position change between consecutive tracking timestamps.*

Secondly, as the replay animation shows on the jumbotron, a QR code appears in the corner of the screen. Fans scan it from their seats, opening a tab on their browser with a single slider and one question tailored to the specific play. The replay will begin after fans are given ample time to guess. For an extra-base hit, fans guess how long it took the lead runner to score or for the batter to reach a given base. Fans lock in their answers as the replay begins.

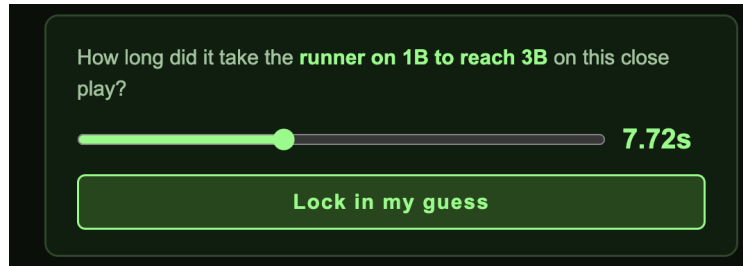


Figure 4. *Mobile interface accessed via QR code during the replay. Fans drag the slider to guess the elapsed time for a key runner to reach the target base and lock in their answer before the replay begins.*

Lastly, the points reveal. When the replay finishes, the actual time appears on the jumbotron alongside the fan's score out of 100. A live leaderboard updates on screen showing how different players in the stadium ranked, turning a single play into a stadium-wide shared moment.



Figure 5. Result screen showing the fan's locked-in guess, the actual time derived from tracking data, their accuracy score out of 100, and a live leaderboard ranking sections of the stadium by guess accuracy.

Three different play types are featured in the following *Race the Replay* demo, each creating a different guessing challenge. The extra-base hit features multiple runners moving simultaneously, making it harder to determine a single runner's time. The wall deflection play adds unpredictability; the ball's path off the wall changes everything as well as the defense's relay time may speed up or slow down the runner. The close play at the plate is the most

action-packed, with a runner caught between fielders and every second mattering to the final answer.

The following links provide a full demonstration of the Race the Replay experience for each of the three featured play types:

Play 1 — Extra-Base Hit: <https://youtu.be/oDswAoUE9bE>

Play 2 — Wall Deflection: <https://youtu.be/gOsaVxqaflY>

Play 3 — Close Play at the Plate: <https://youtu.be/YvAnWrYBWaw>

A full scoreboard operator script and fan experience walkthrough are provided in Appendix B.

Conclusion

Race the Replay demonstrates that the information needed for meaningful fan engagement already exists inside Minor League Baseball today. The same player and ball tracking data used by front office analysts to evaluate sprint speed, defensive positioning, and baserunning decisions can be transformed into an experience that pulls fans deeper into the game they are watching. What makes *Race the Replay* different from scoreboard content is not the data itself, but the relationship it creates between the fan and the play. *Race the Replay* actually asks fans to engage with the material and make a prediction. That change from passive consumption to active participation is what distinguishes it from other jumbotron experiences.

The current game was built entirely within the constraints of an anonymized dataset, and those constraints transformed the way the project looks right now in many ways. In a real deployment with full player data, the experience would be even richer. Each runner would be replaced by the player's actual headshot, making it immediately obvious to fans which player they are tracking. Jersey numbers would appear alongside each player on the field, and the question itself would reference the player by name: "How long did it take James Wood to score on his inside-the-park homerun?" rather than a generic example as in the demo. Team colors would replace the neutral color scheme, enhancing the replay visually. Stadium geometry, absent from this dataset, would allow the animation to reflect the actual dimensions and special wall contours of the home ballpark.

Overall, *Race the Replay* suggests that the next movement for fan engagement in baseball is not more information on the scoreboard but better questions behind how that information is shared. The data exists. The infrastructure exists. What is missing is a product that turns tracking into a moment fans want to be part of, which is exactly what *Race the Replay* is designed to do.

Acknowledgments

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during the play verification phase of this project. Additional thanks to SMT for providing the data, hosting the virtual office hours, and organizing the tutorials that made this project possible.

References/Citations

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Appendix

Appendix A: Technical Methodology- Base Coordinate Calculation

The SMT coordinate system places home plate at the origin (0, 0), with the x-axis extending toward first base and the y-axis towards second base. Using the standard baseball 90-foot base path distance, the four base coordinates were calculated as:

- Home plate: (0, 0)
- First base: (63.63, 63.63)
- Second base: (0, 127.27)
- Third base: (-63.63, 63.63)

These values are derived from the fact that the base paths form a square rotated 45 degrees, placing each base at $90/\sqrt{2} \approx 63.63$ feet along both axes from home plate.

Base Touch Detection:

At every timestamp, the distance between each runner's position (x, y) and each base coordinate (bx, by) was computed:

$$d = \sqrt{(x - bx)^2 + (y - by)^2}$$

The moment this distance fell below 10 feet was recorded as the runner's arrival time. This threshold accounts for variance in sensor readings. Elapsed time was then calculated from the ball-hit timestamp to produce the final time-to-base value used in the game.

Player Speed Calculation:

Speed was derived from consecutive position readings:

$$\text{speed (ft/s)} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} / ((t_2 - t_1) / 1000)$$

Values were normalized to a 0-to-1 scale using a ceiling of 35 ft/s, the upper range of sprint speed in professional baseball:

$$\text{speed_norm} = \min(\text{speed} / 35, 1.0)$$

This normalized value drives the velocity trail color, mapping from blue (0.0) through yellow (0.5) to red (1.0).

Play Selection Filters:

Plays were evaluated using three filters:

1. Throw count: total throws (`ball_eventcode = 3`) during the play. Two or more indicates a multi-fielder defensive sequence.
2. Wall deflection flag: whether a ball deflection off wall event (`ball_eventcode = 10`) occurred.
3. Runners: confirmed via the lineups table, checking whether `on_1b`, `on_2b`, or `on_3b` was populated.

Plays were ranked by throw count descending, with duration as a tiebreaker. Raw duration was excluded as a primary filter after several of the longest plays proved to be timing irregularities rather than extended on-field action.

Appendix B: App Documentation

Scoreboard Operator Script

Race the Replay is triggered manually by a scoreboard operator following a significant play earlier in the game. The recommended timing is during a natural break in the sixth or seventh inning. The operator selects the appropriate play file from the three pre-loaded options and initiates the experience. A QR code should appear in the corner of the jumbotron display for the duration of the replay, giving fans time to scan and submit their guess before the animation ends. The operator advances to the reveal screen once the animation completes.

Fan Experience Walkthrough

1. The jumbotron displays the Race the Replay animation. Fans see an animated top-down field view reconstructing the selected play from tracking data.
2. A QR code appears in the corner of the screen. Fans scan it from their seats, opening the mobile interface in their browser — no app download required.
3. Fans read the play-specific question, drag the slider to their guess, and tap "Lock in my guess" before the replay.

4. When the animation finishes, the jumbotron displays the actual time, each fan's personal score out of 100, and a live leaderboard showing how the stadium ranked.

Technical Requirements for Live Deployment

The current prototype runs as a local HTML file served alongside three JSON data files. For live deployment the following would be required:

- A web server hosting the scoreboard app and play data files
- A mobile-accessible URL tied to the QR code displayed on the jumbotron
- A live backend database to collect fan submissions and power the real-time leaderboard
- A unique QR code generated per game and per play to prevent cross-game score contamination

File Structure

The Race the Replay prototype consists of four files:

- scoreboard.html — the full scoreboard and fan interface
- play1_extra_base.json — tracking data for the extra-base hit play
- play2_wall_deflection.json — tracking data for the wall deflection play
- play3_close_play.json — tracking data for the close play at the plate

Each JSON file contains ball positions, runner positions with normalized speed values, defender positions, and the correct time-to-base answers for the guessing game.

Appendix C: Analysis Code

The full analysis was conducted in R using the `arrow`, `tidyverse`, and `jsonlite` packages. The code is organized into two files:

- `SMT_Setup.R` — loads all data connections and defines helper functions provided by SMT
- `SMT_Analysis.Rmd` — contains the full analysis pipeline including play selection, time-to-base calculation, speed normalization, and JSON export

The scoreboard application is built as a single self-contained HTML and JavaScript file (`scoreboard.html`) which loads the three play JSON files to power the animation and fan guessing interface. The scoreboard application was built in HTML and JavaScript. Key resources used in development include the MDN Canvas API documentation (developer.mozilla.org) and the `requestAnimationFrame` API for frame-by-frame animation.

All code is submitted separately as part of the required submission package.

